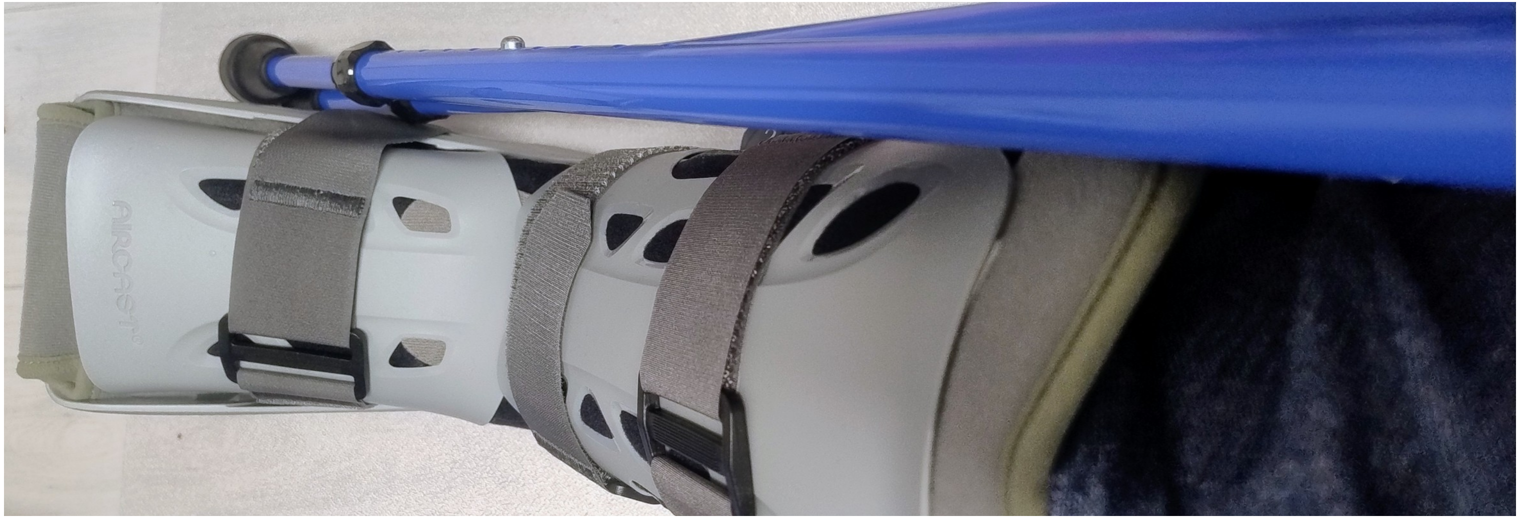
Longterm latency monitoring of Linux with PREEMPT_RT

Jan Altenberg,
Open Source Automation Development Lab (OSADL) eG

ELISA Workshop: London, UK
June 9-11, 2026
Co-hosted with Canonical

Sorry for not being there...



AI generated content (Canva Pro)

$ whoami

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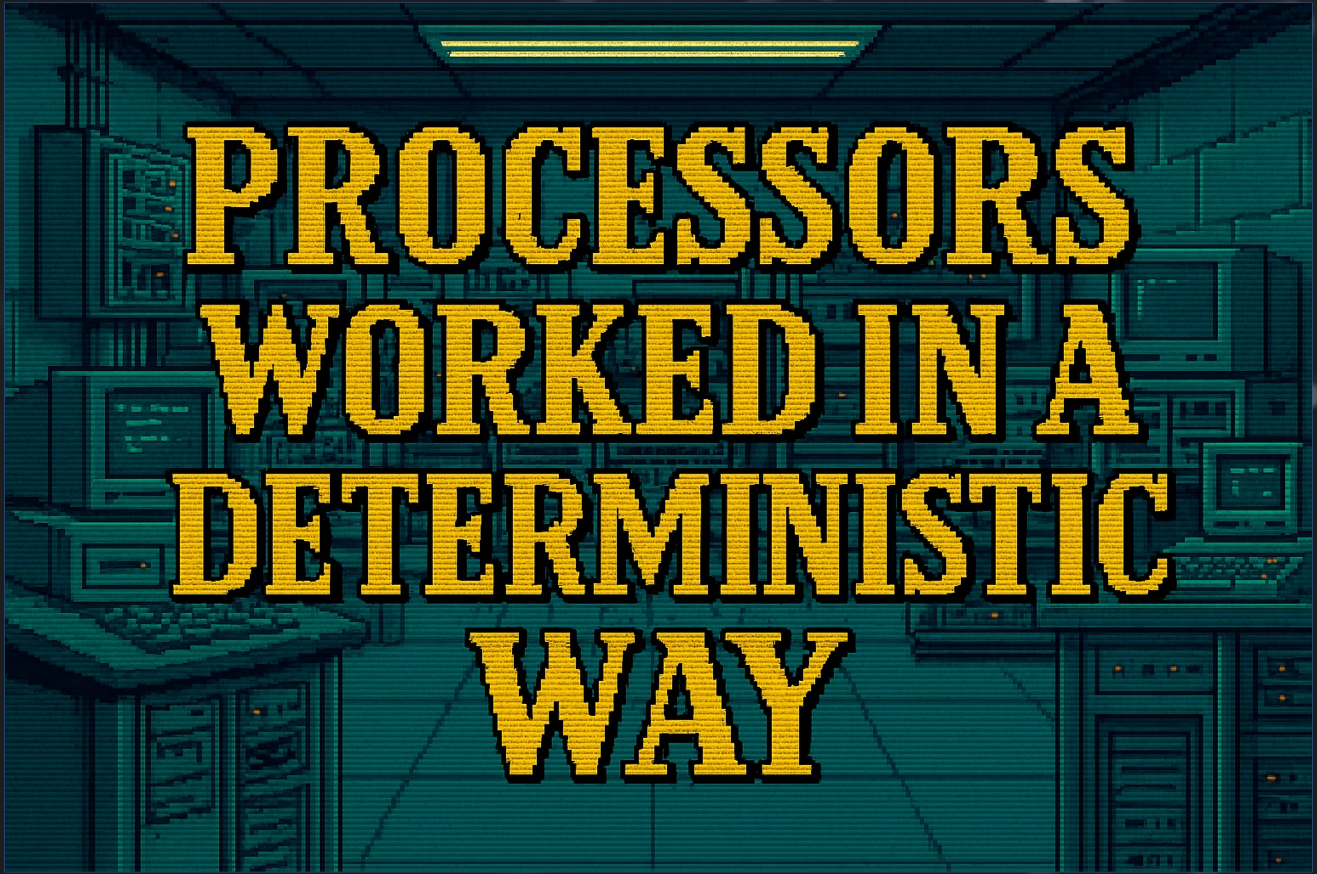

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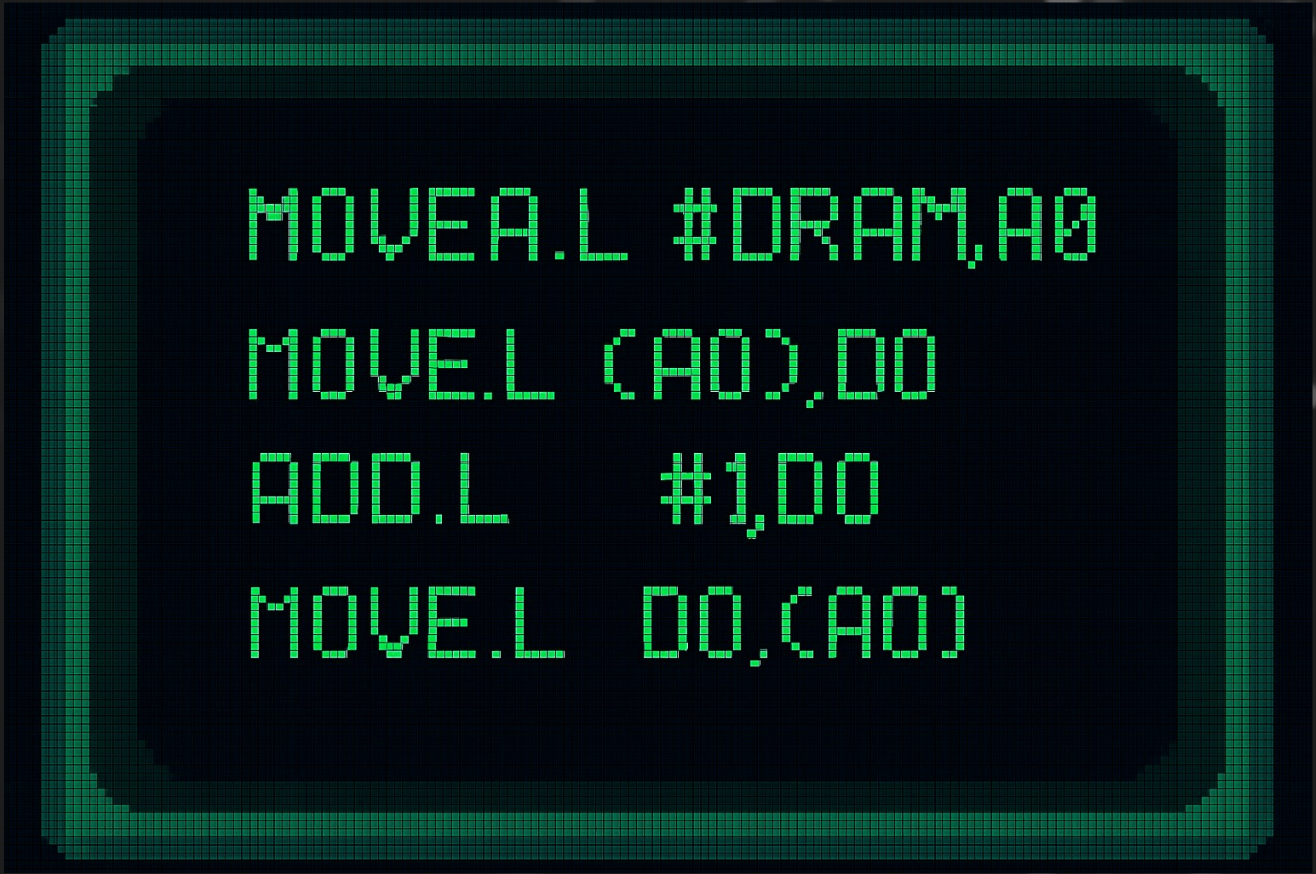

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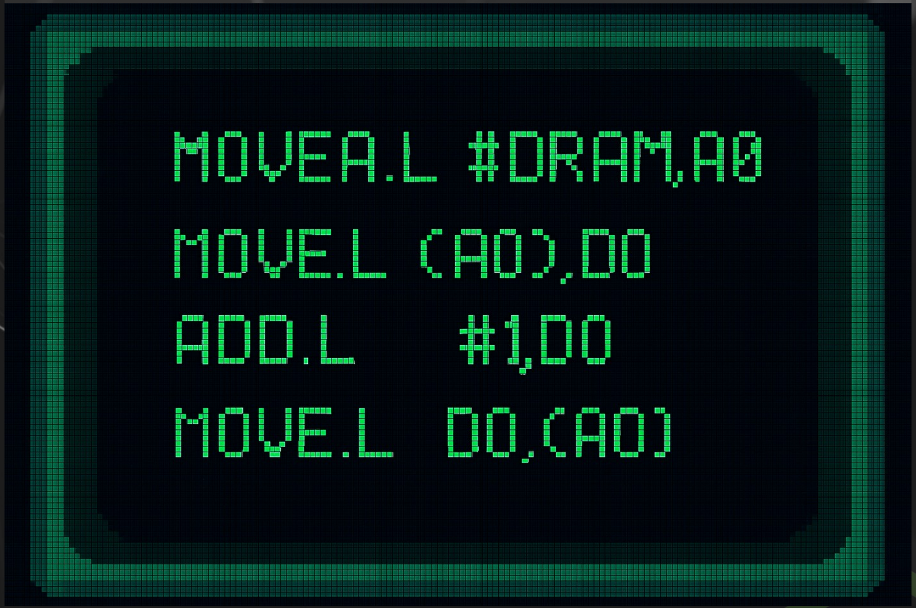
1979, Motorola MC68000 @ 8 MHz (600 Dhrystones)



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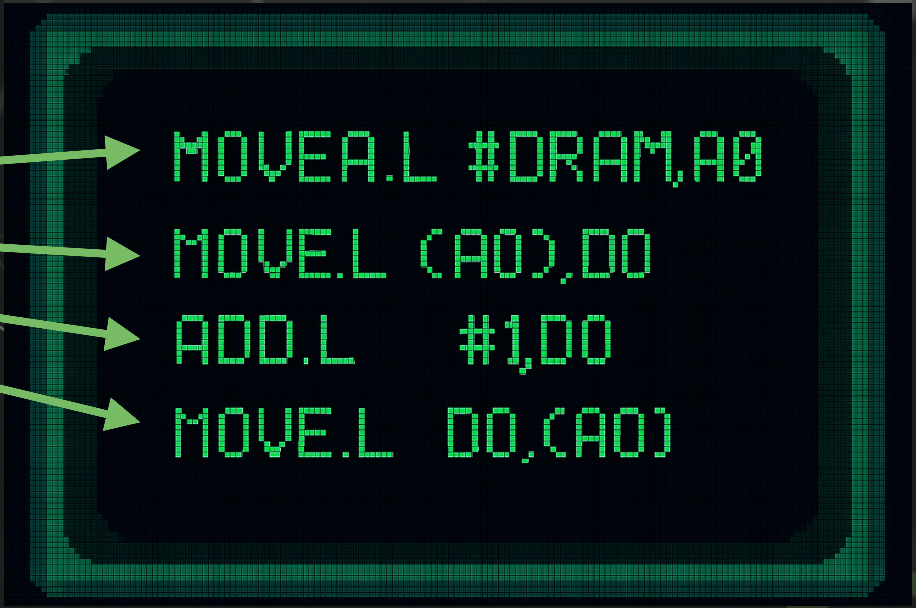
1979, Motorola MC68000 @ 8 MHz
(600 Dhrystones)

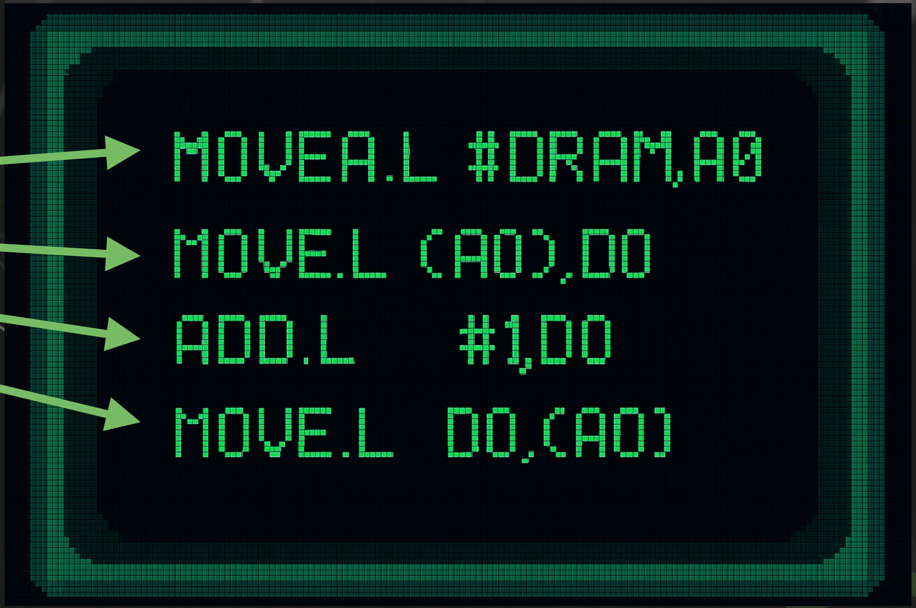
Fetch instructions from memory and execute them.



AI generated content (Copilot)

1979, Motorola MC68000 @ 8 MHz
(600 Dhrystones)

Fetch instructions from memory and execute them.
Duration = <u>**56**</u> processor cycles



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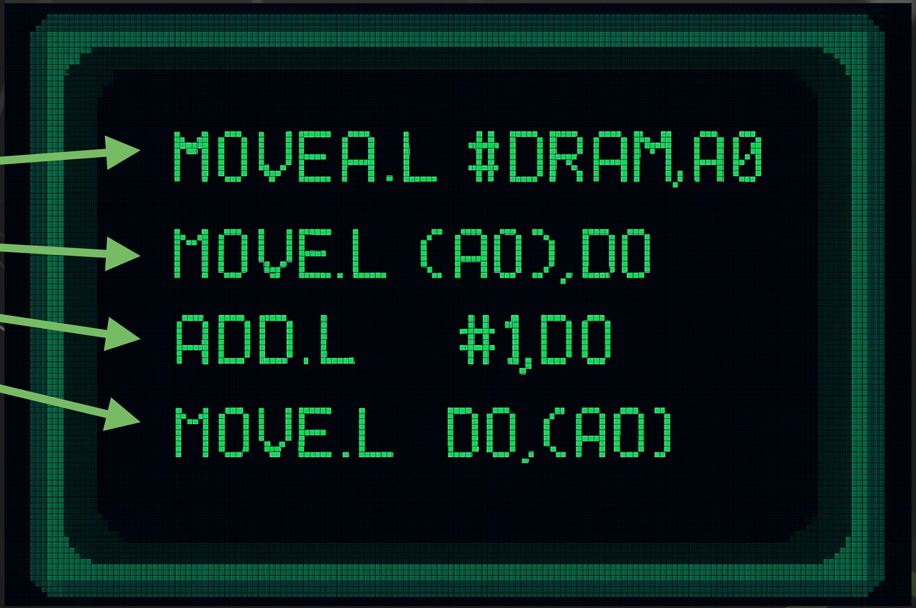
WORKSHOP

1979, Motorola MC68000 @ 8 MHz (600 Dhrystones)

Fetch instructions from memory and execute them. Duration = **56** processor cycles

SLOW.



AI generated content (Copilot)

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WORKSHOP

1979, Motorola MC68000 @ 8 MHz
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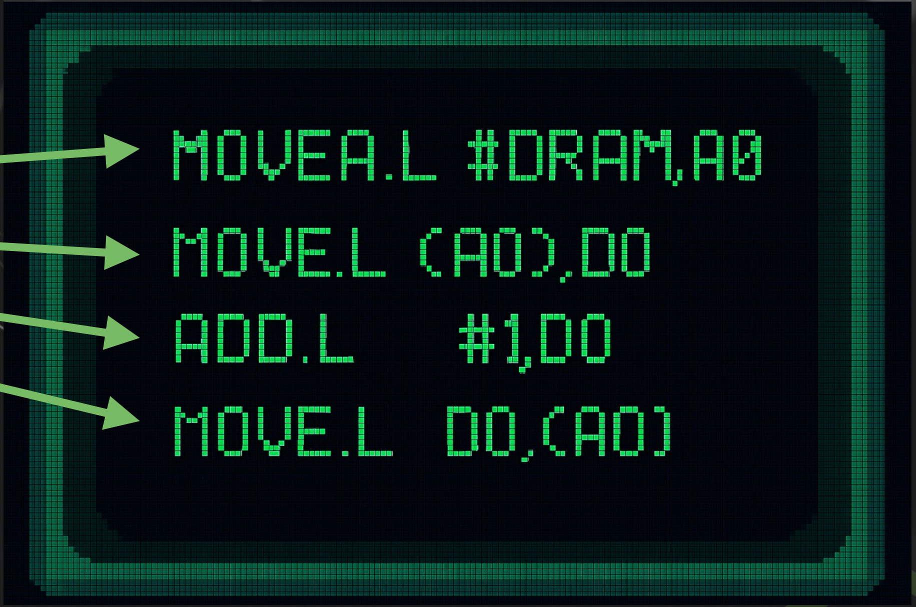
Fetch instructions from memory and execute them. Duration = **<u>56</u>** processor cycles



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SLOW, BUT DETERMINISTIC

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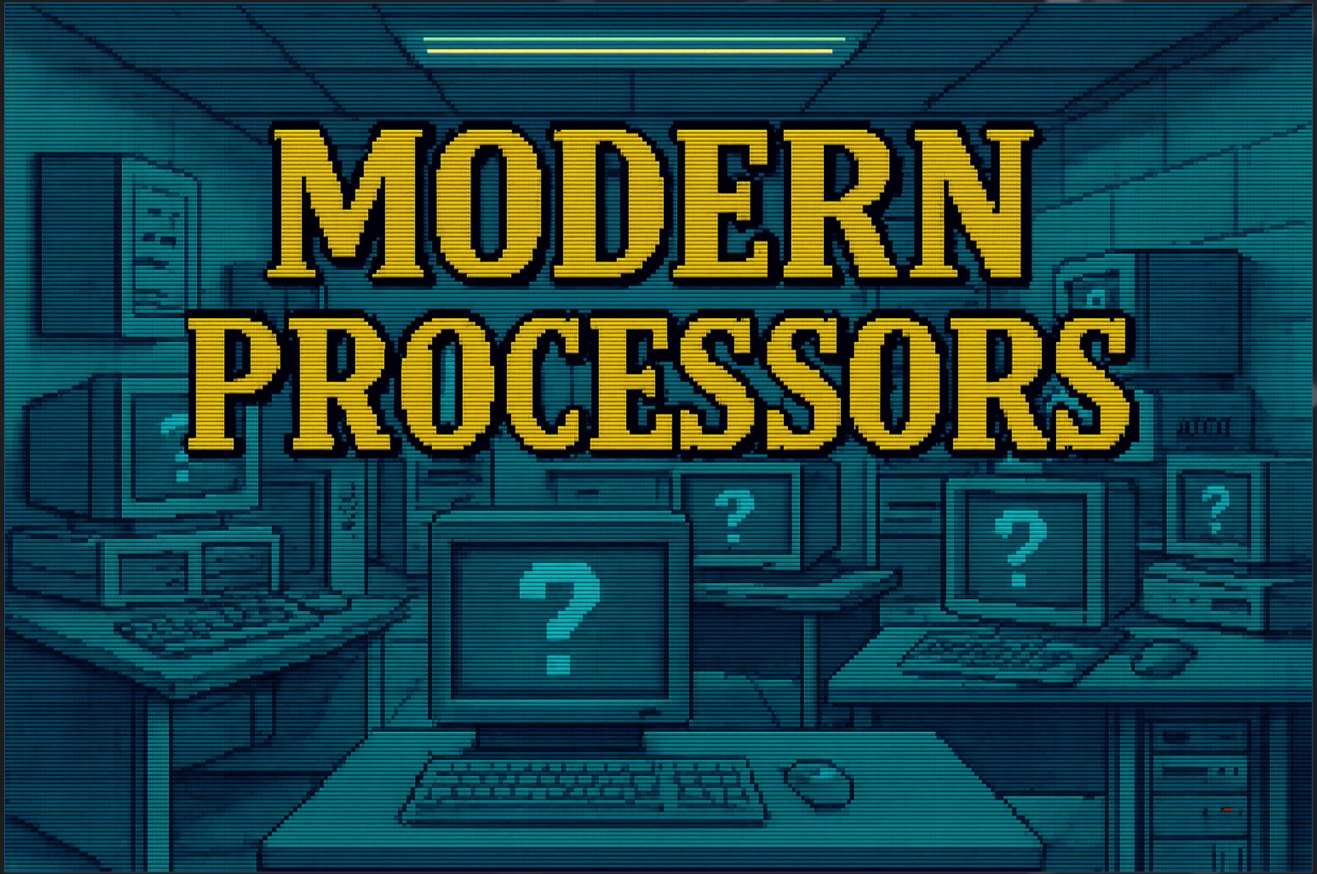

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```
mov  dram,eax
mov  eax,-4(ebp)
add  $1, -4(bp)
mov  -4(ebp),eax
mov  eax,dram
```

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WORKSHOP

2009, x86 processor @ 3 GHz
(12.000.000 Dhrystones)

Fetch instructions from memory and execute them.

```
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*AI generated content (Copilot)

ELISA Enabling Linux in Safety Applications WORKSHOP

2009, x86 processor @ 3 GHz
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Fetch instructions from memory and execute them. Duration = **<u>????</u>** processor cycles

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mov  dram,eax
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Instruction not in cache

2009, x86 processor @ 3 GHz
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```

*AI generated content (Copilot)

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Instruction not in cache

Data not in cache/no free cache lines

2009, x86 processor @ 3 GHz
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Instruction not in cache

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System Management Interrupt

2009, x86 processor @ 3 GHz
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mov  dram,eax
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Instruction not in cache

Data not in cache/no free cache lines

System Management Interrupt

Instruction may need to be emulated (microcode patch)

2009, x86 processor @ 3 GHz
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mov  dram,eax
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* AI generated content (Copilot)

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WORKSHOP

2009, x86 processor @ 3 GHz
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add  $1, -4(bp)
mov  -4(ebp),eax
mov  eax,dram
```

*AI generated content (Copilot)

FAST, BUT RANDOM

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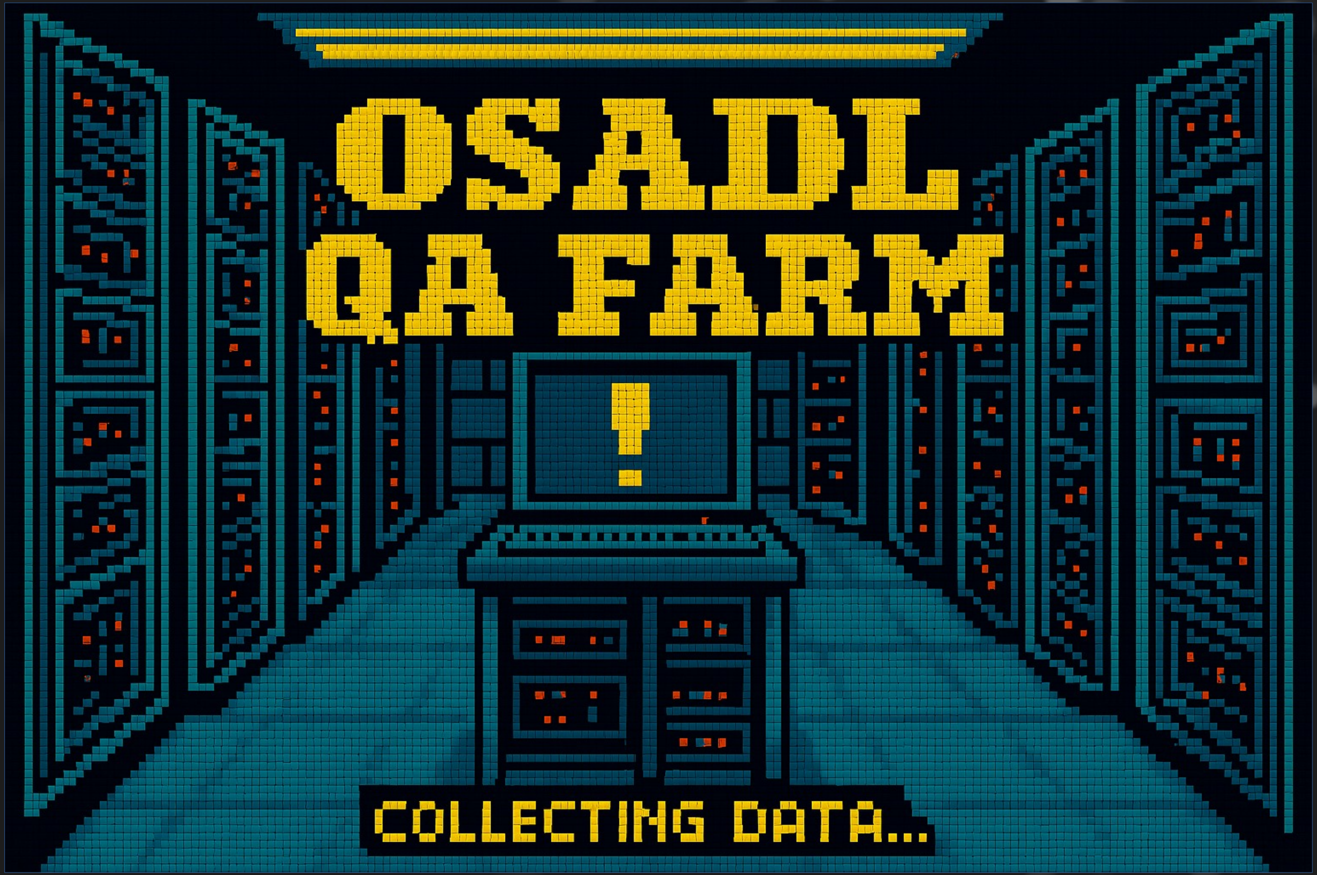
Challenge

- Transition from static code analysis to dynamic system analysis
- Transition from theoretical approach to empirical testing
- Transition from one paradigm to another

Evaluation methods

- External measurement with simulation (e.g. function generator)
- Internal latency registration (e.g. OSADL patches, RTLA)
- Internal measurement with simulation (e.g. cyclictest)



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OSADL QA Farm

https://www.osadl.org/QA

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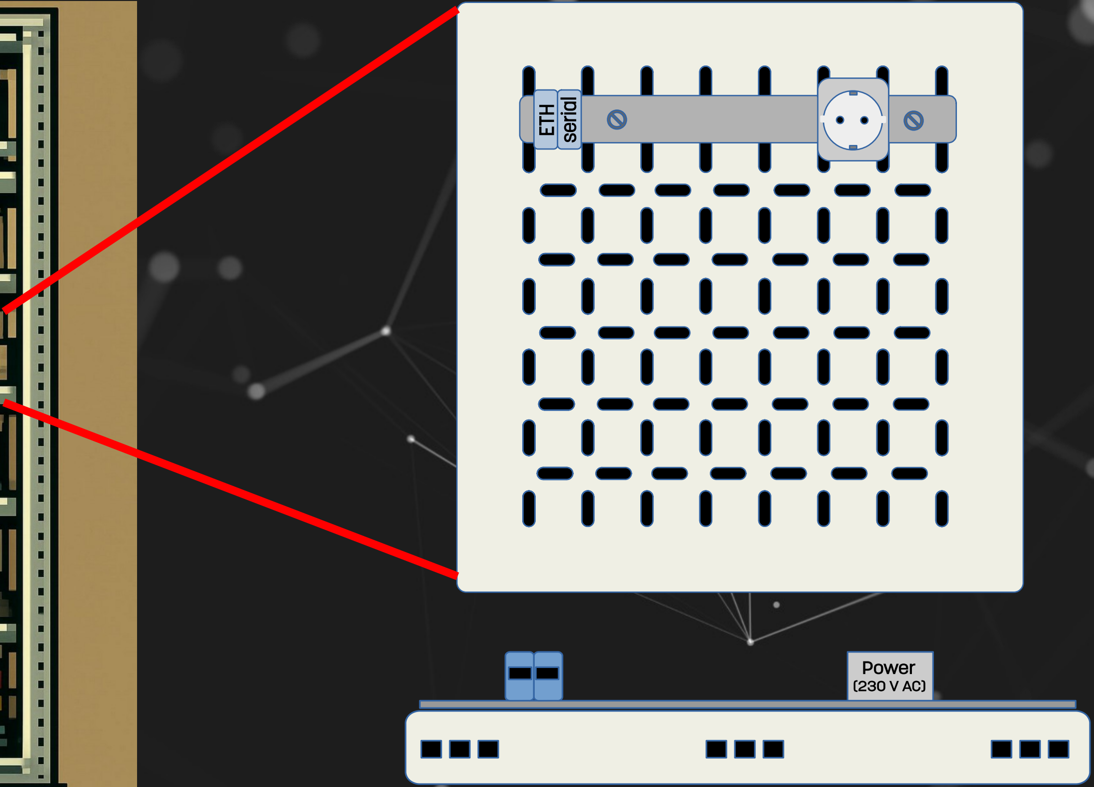

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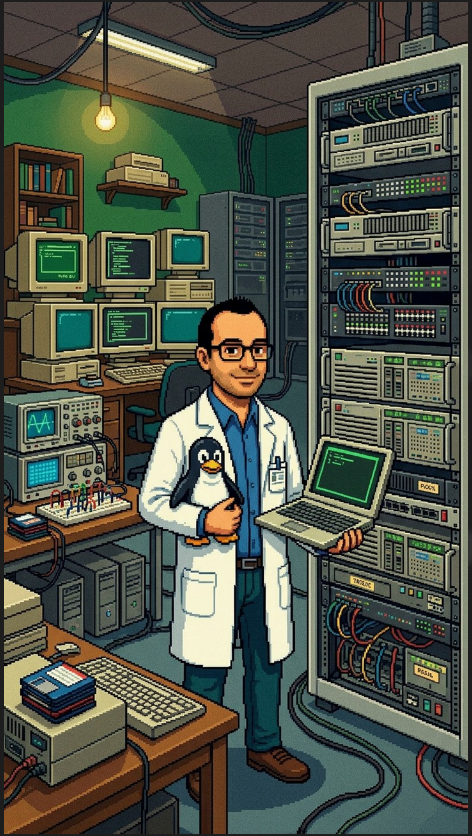
Data being recorded:



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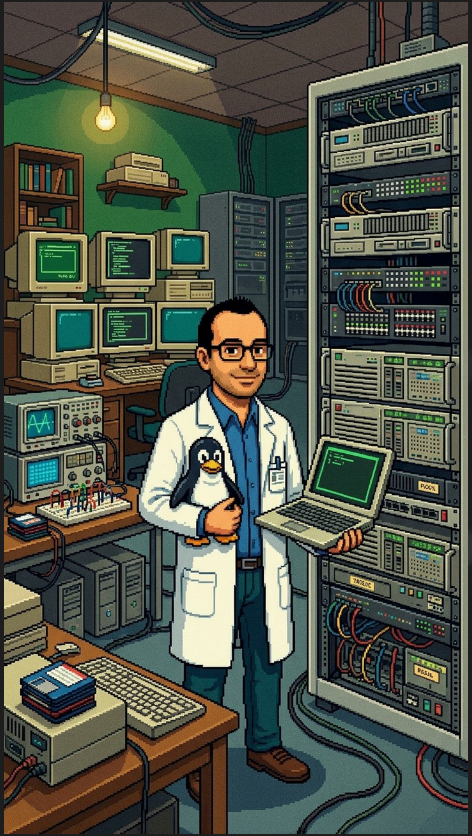

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- Static data (vendor / board, kernel version, kernel configuration, ...)

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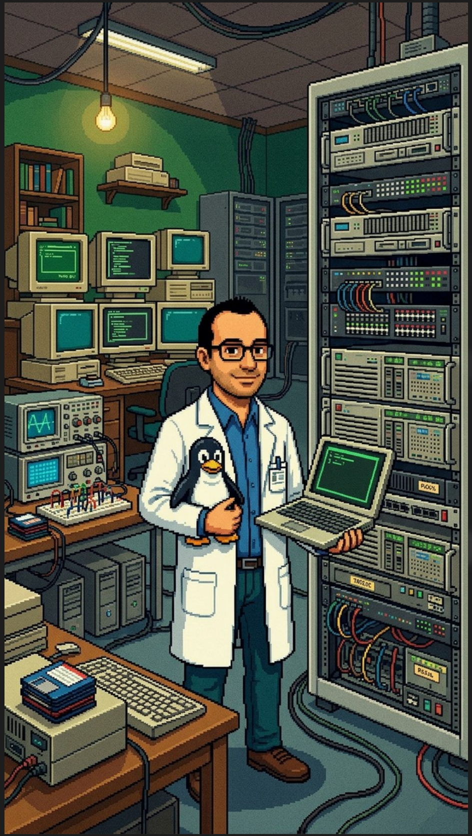

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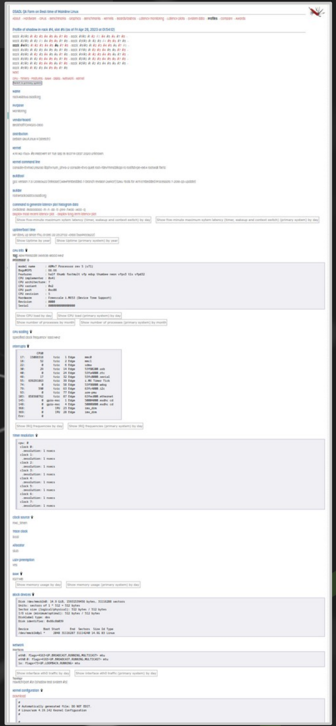

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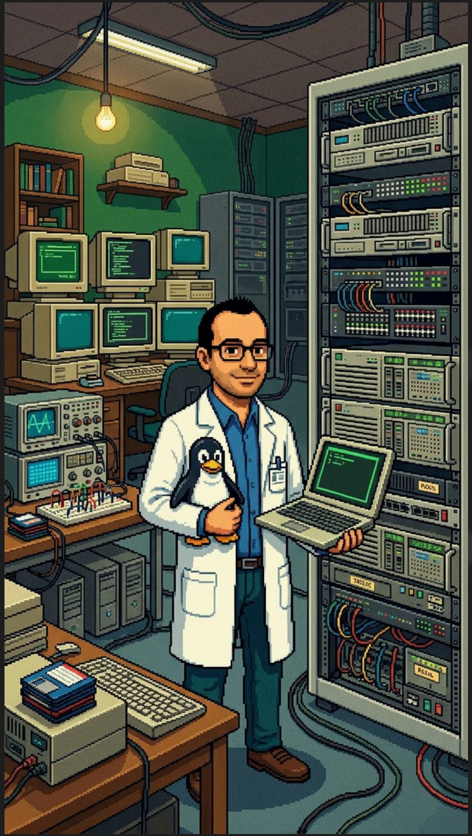

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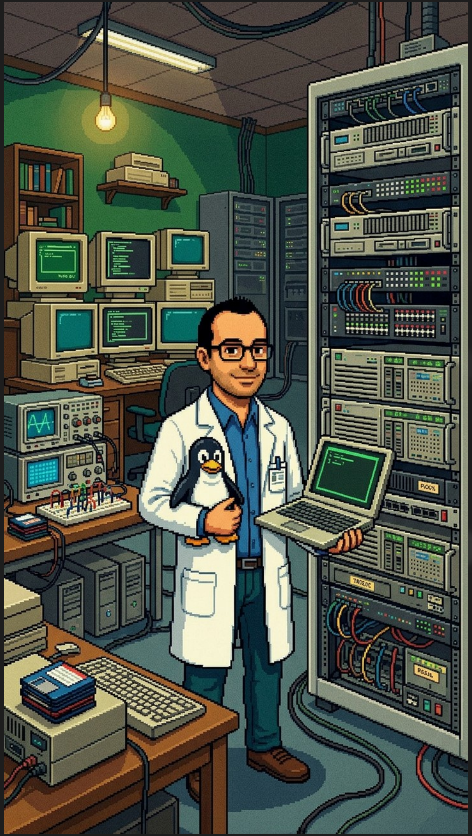
Data being recorded:

- Static data (vendor / board, kernel version, kernel configuration, ...)
- Continuous measurements (CPU load, memory consumption, temperatures, uptime, ...)



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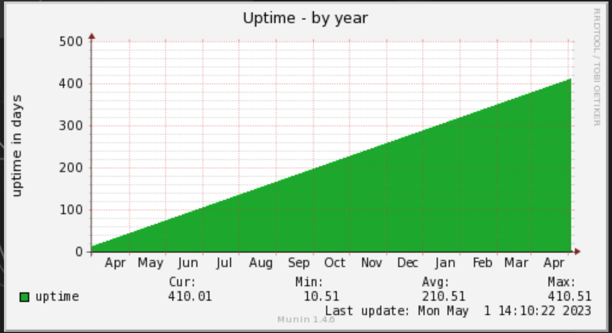

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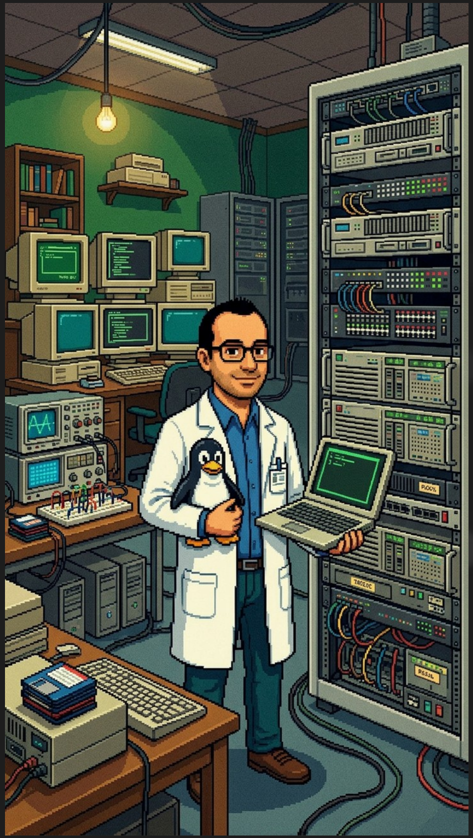

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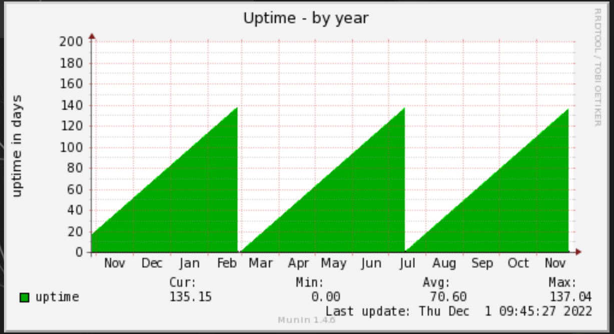

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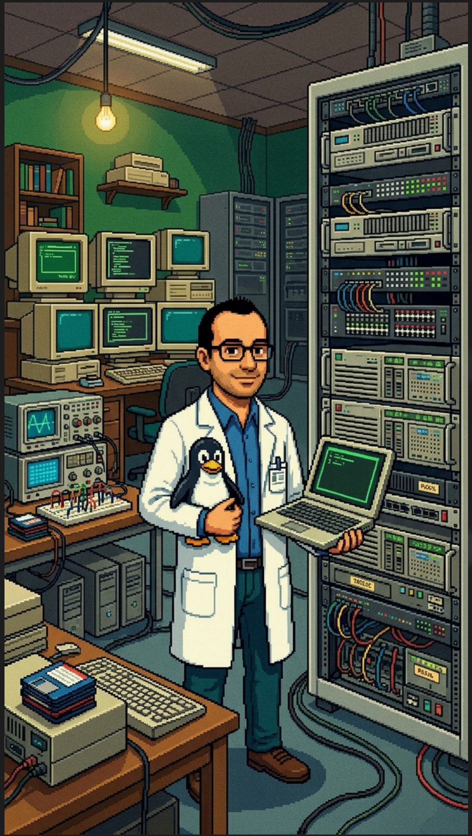

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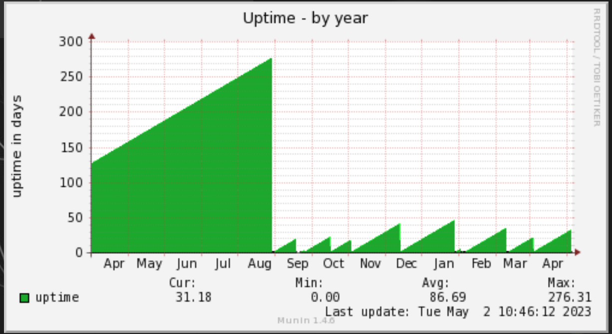

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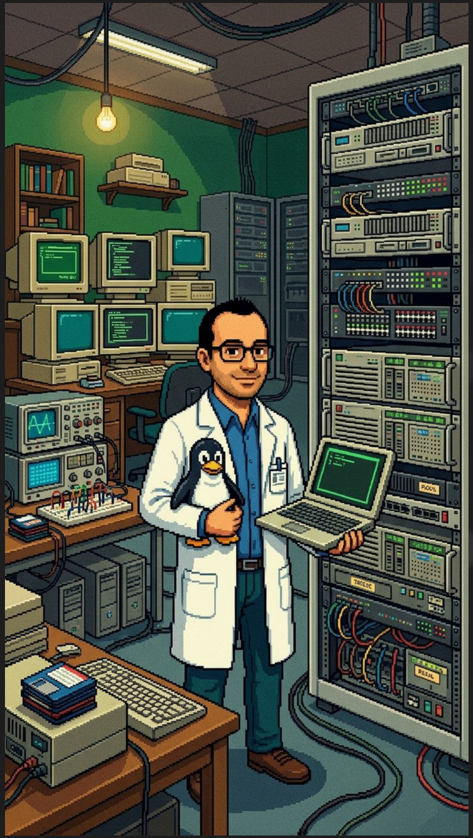

AI generated content (Canva Pro)

Data being recorded:

- Static data (vendor / board, kernel version, kernel configuration, ...)
- Continuous measurements (CPU load, memory consumption, temperatures, uptime)
- Latency measurements

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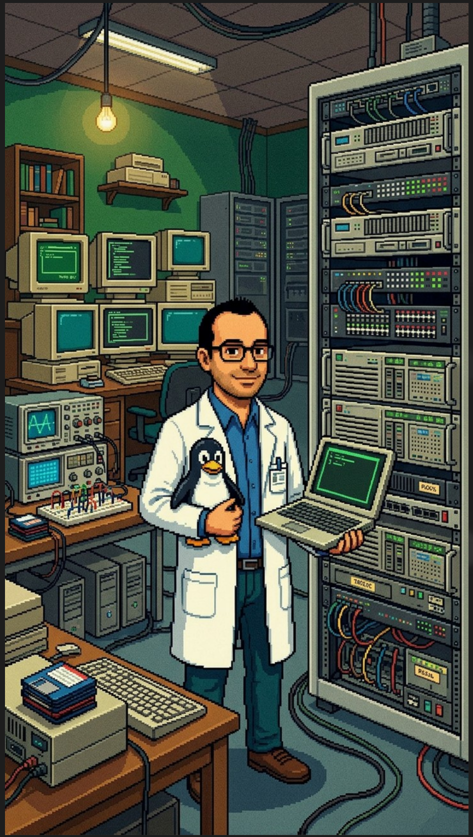

AI generated content (Canva Pro)

Data being recorded:

- Static data (vendor / board, kernel version, kernel configuration, ...)
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- **Latency measurements**

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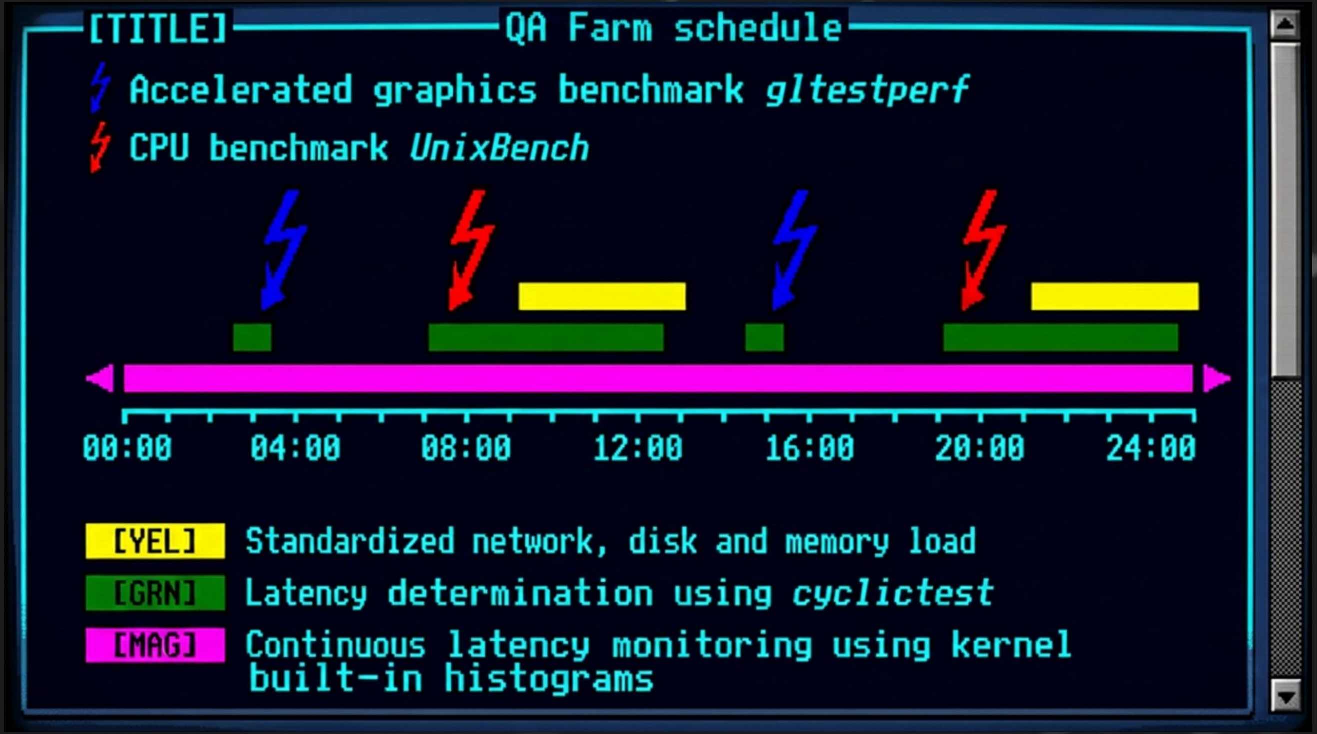
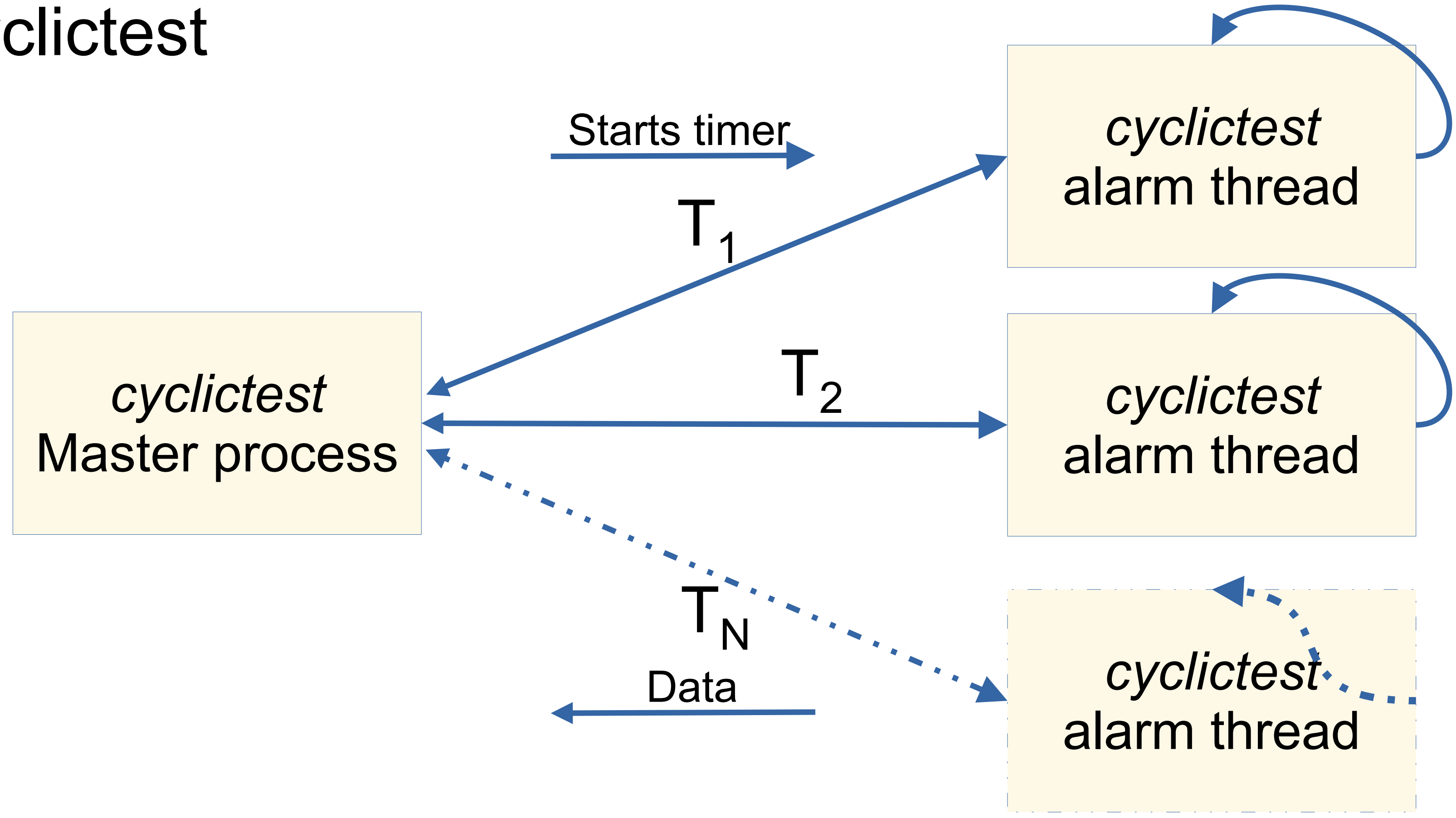


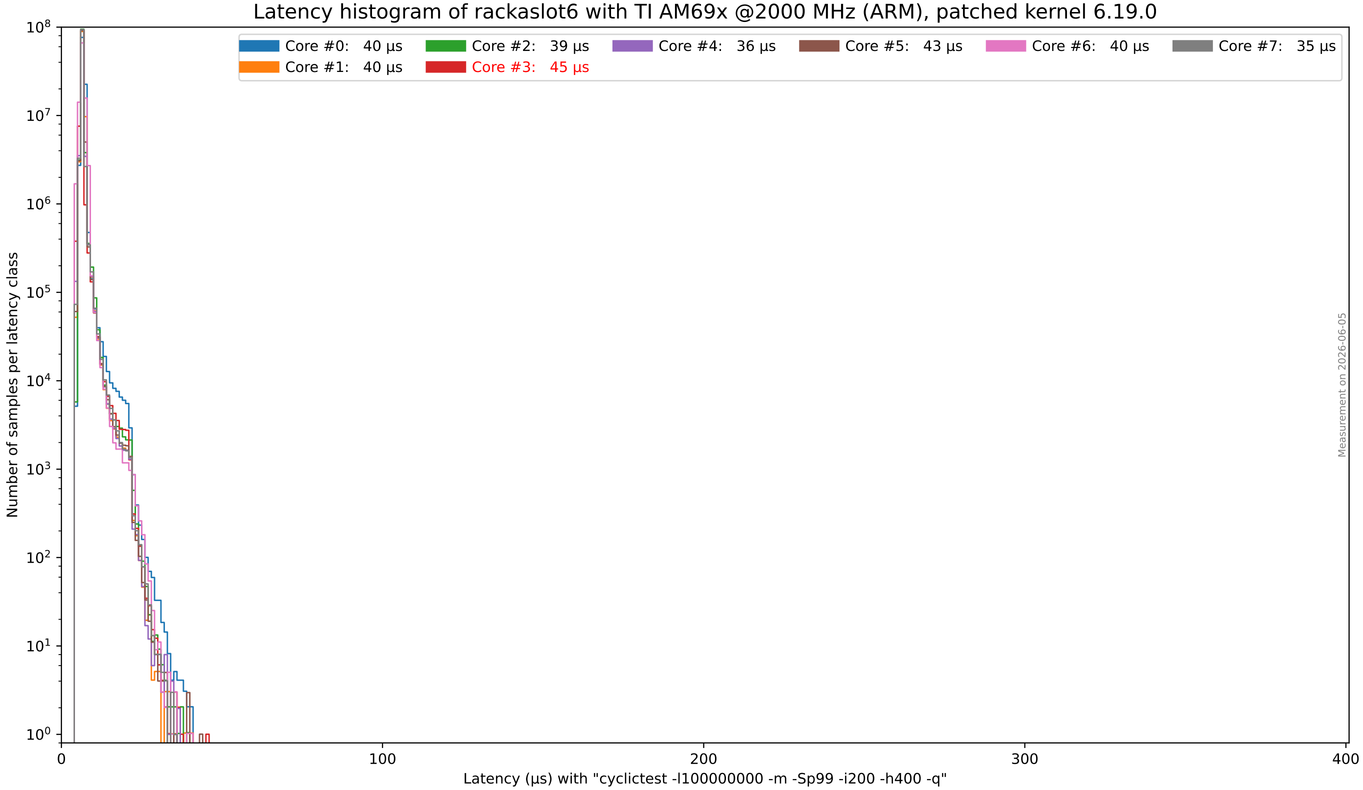
AI generated content (Canva Pro)

cyclictest



Starts timer
cyclictest
alarm thread
T1
cyclictest
Master process
T2
cyclictest
alarm thread
TN
Data
cyclictest
alarm thread

cyclictest

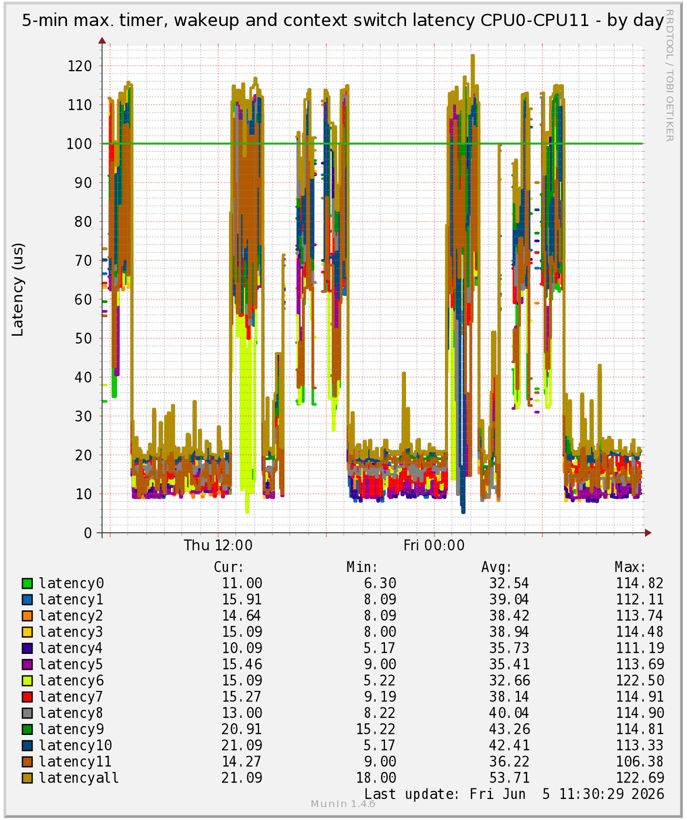


Latency histogram of rackaslot6 with TI AM69x @2000 MHz (ARM), patched kernel 6.19.0
Core #0: 40 µs
Core #1: 40 µs
Core #2: 39 µs
Core #3: 45 µs
Core #4: 36 µs
Core #5: 43 µs
Core #6: 40 µs
Core #7: 35 µs
Number of samples per latency class
Latency (µs) with "cyclictest -l100000000 -m -Sp99 -i200 -h400 -q"
Measurement on 2026-06-05

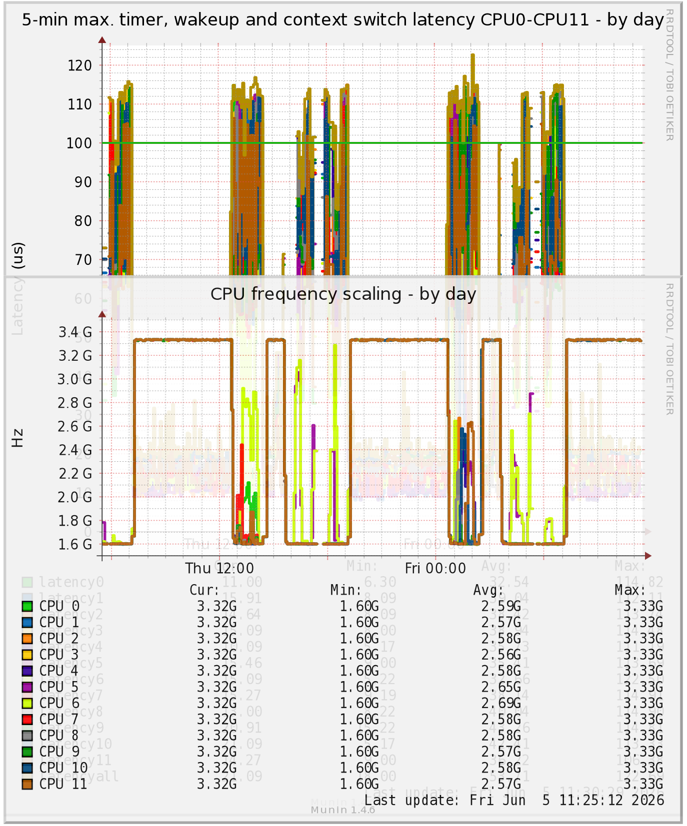
Latency histograms

https://www.osadl.org/?id=2945

Latency histograms



Latency histograms



Latency histograms

```
# cyclictest -m -S -i200 -d0 -p99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 0.05 0.06 0.02 1/131 493

T: 0 (  474) P:99 I:200 C:  81314 Min:      6 Act:    9 Avg:    8 Max:    4952
T: 1 (  475) P:99 I:200 C:  81243 Min:      6 Act:    9 Avg:    8 Max:    4987
T: 2 (  476) P:99 I:200 C:  81191 Min:      6 Act:    9 Avg:    8 Max:    4880
T: 3 (  477) P:99 I:200 C:  81124 Min:      6 Act:   11 Avg:    8 Max:    4886
```

Latency histograms

```
# cd /sys/kernel/debug/latency_hist/timerandwakeup

# cat max_latency-CPU*

546 99 4988 (4964,0) cyclictest <- 552 -21 mklatency 440.676730
547 99 4875 (4849,0) cyclictest <- 553 -21 mklatency 440.687647
548 99 4929 (4909,0) cyclictest <- 554 -21 mklatency 440.697379
549 99 4874 (4849,0) cyclictest <- 555 -21 mklatency 440.707655
```

Latency histograms

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549 99 4874 (4849,0) cyclictest <- 555 -21 mklatency 440.707655
```

Victim

Latency histograms

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```

Priority

Latency histograms

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```

Latency (us)

Latency histograms

```
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```

Culprit

Latency histograms

```
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Priority

Latency histograms

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```
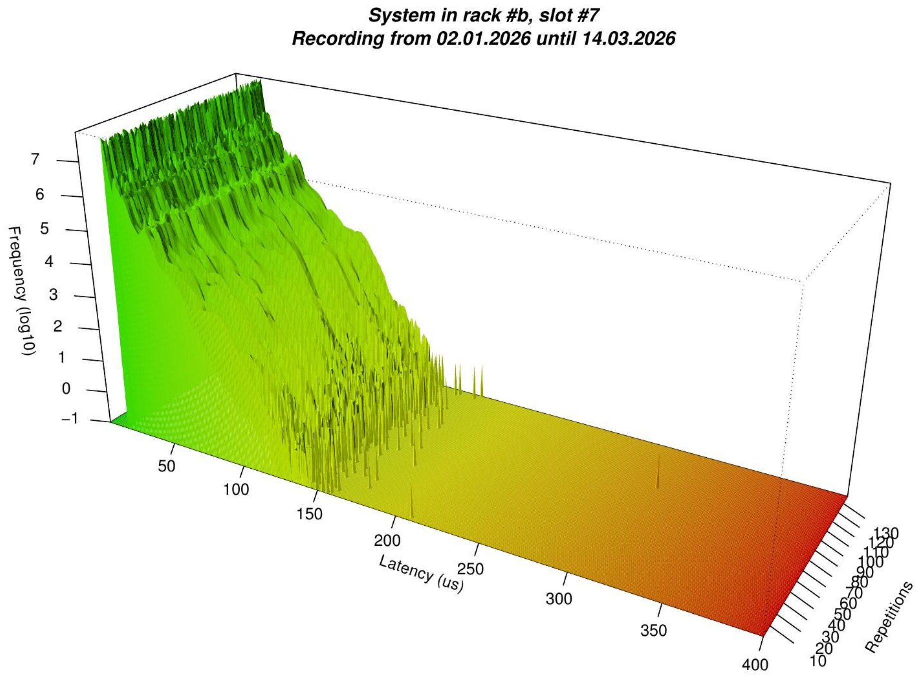

Timestamp

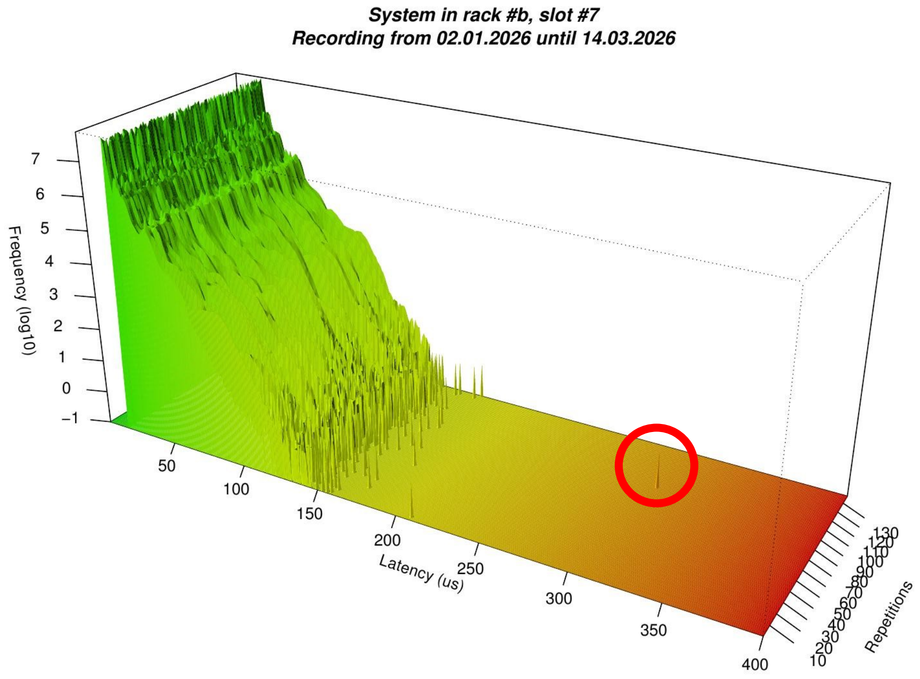
Latency histograms

System rack6slot2.osadl.org (updated Fri Jun 05, 2026 00:43:26)									
Delayed (victim)					Switcher (culprit)			Timestamp	CPU
PID	Prio	Total latency (µs)	T*(,W**) latency (µs)	Cmd	PID	Prio	Cmd		
28785	2	48	8,8	sleep0	0	-21	swapper/0	19:04:20	0
29068	2	43	9,8	sleep1	0	-21	swapper/1	19:07:20	1
29151	99	34	32,1	cyclictest	1687	50	irq/16-i915	00:28:09	1
29150	99	34	33,0	cyclictest	2466	50	irq/16-enp2s0f0	23:17:07	0
29150	99	34	33,0	cyclictest	2466	50	irq/16-enp2s0f0	21:06:35	0
29150	99	34	33,0	cyclictest	2466	50	irq/16-enp2s0f0	21:01:00	0
29150	99	34	33,0	cyclictest	2466	50	irq/16-enp2s0f0	20:55:55	0
29150	99	34	33,0	cyclictest	2466	50	irq/16-enp2s0f0	19:27:13	0
29150	99	34	33,0	cyclictest	2466	50	irq/16-enp2s0f0	19:15:42	0
29150	99	34	32,1	cyclictest	2466	50	irq/16-enp2s0f0	23:29:38	0
29150	99	34	32,1	cyclictest	2466	50	irq/16-enp2s0f0	22:48:24	0
29150	99	34	32,1	cyclictest	2466	50	irq/16-enp2s0f0	21:52:15	0
29150	99	34	32,1	cyclictest	2466	50	irq/16-enp2s0f0	20:09:06	0
29151	99	33	32,0	cyclictest	2466	50	irq/16-enp2s0f0	21:44:09	1
29151	99	33	32,0	cyclictest	2466	50	irq/16-enp2s0f0	21:28:22	1
29151	99	33	32,0	cyclictest	2466	50	irq/16-enp2s0f0	21:16:13	1
29151	99	33	32,0	cyclictest	2466	50	irq/16-enp2s0f0	20:45:59	1

Long-term latency plots



Long-term latency plots



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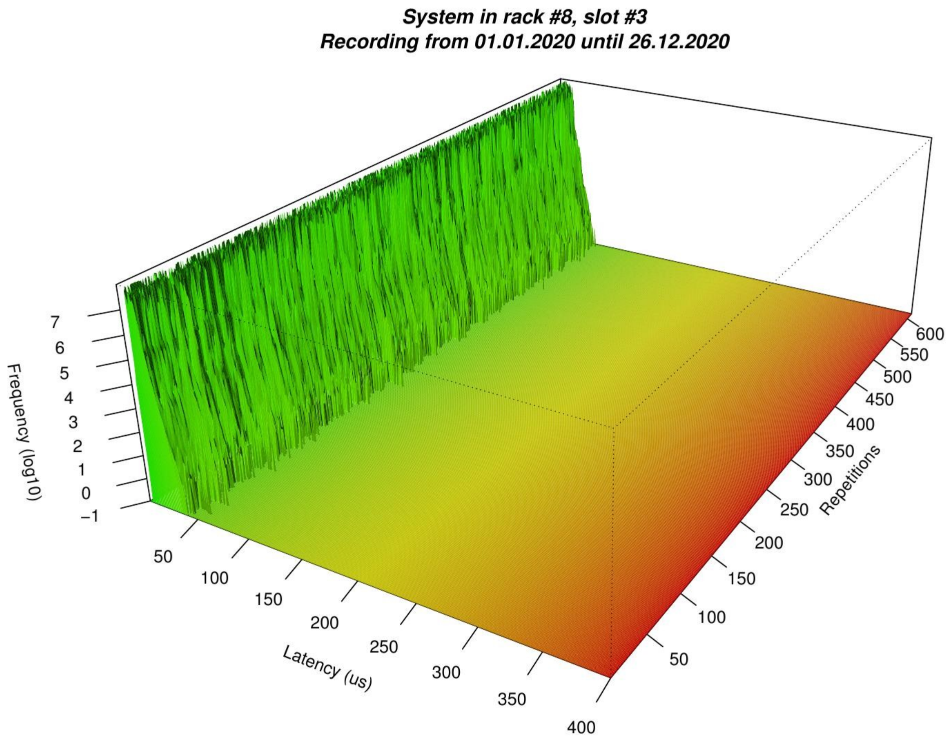

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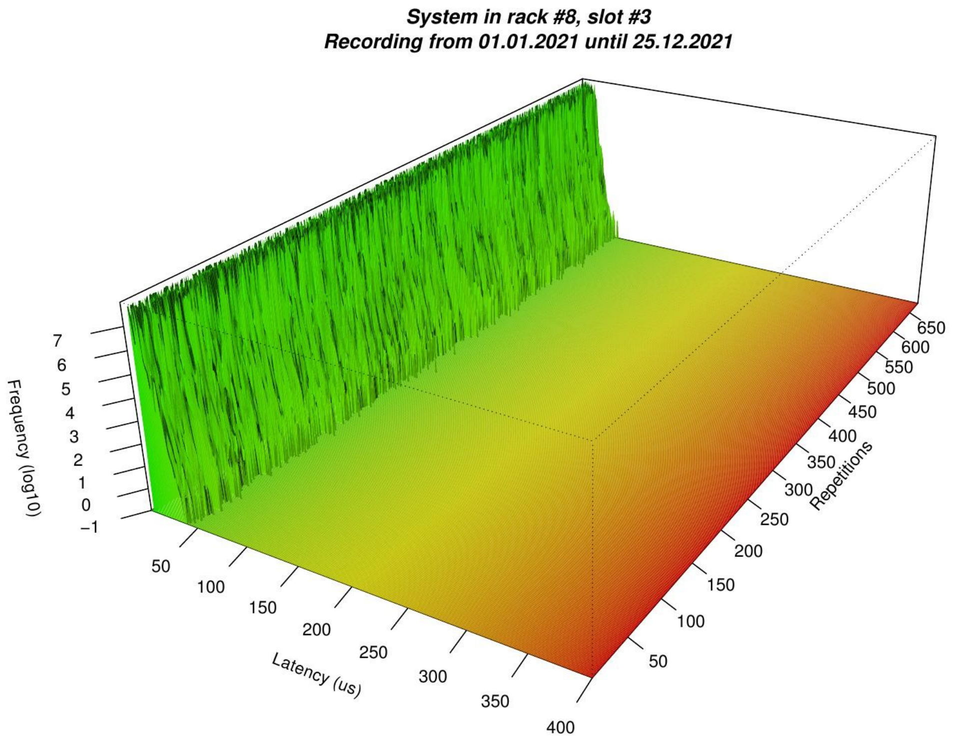

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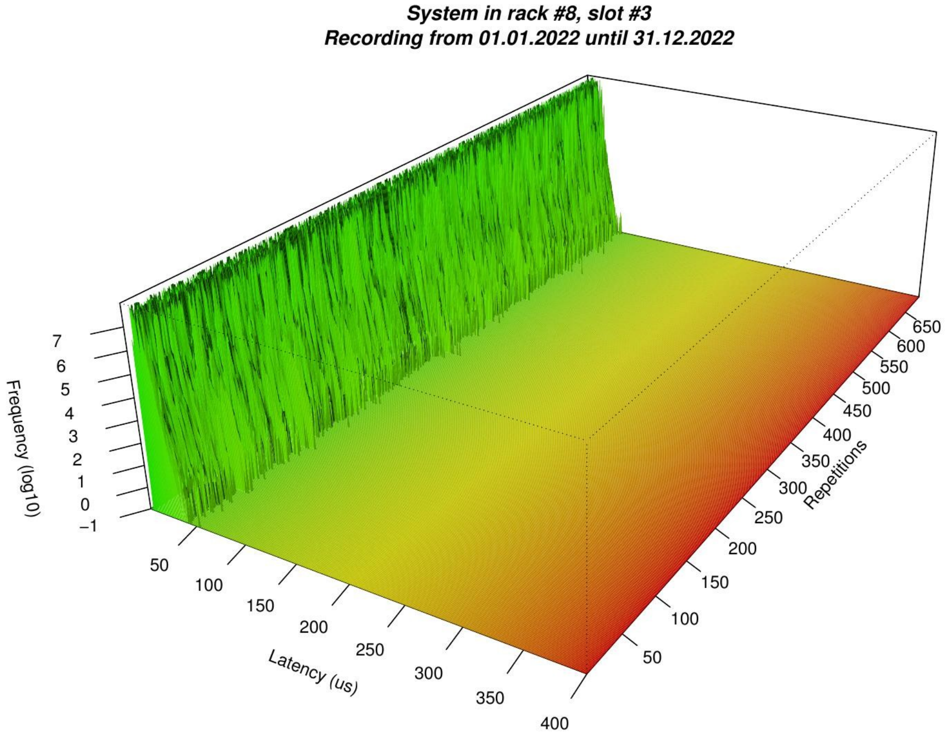

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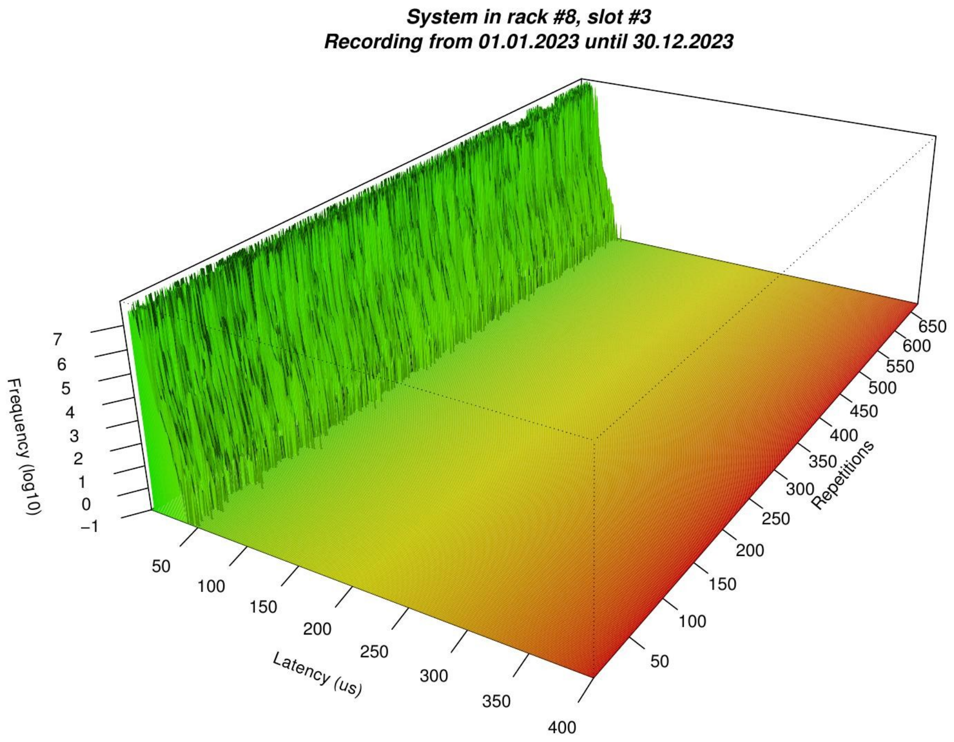

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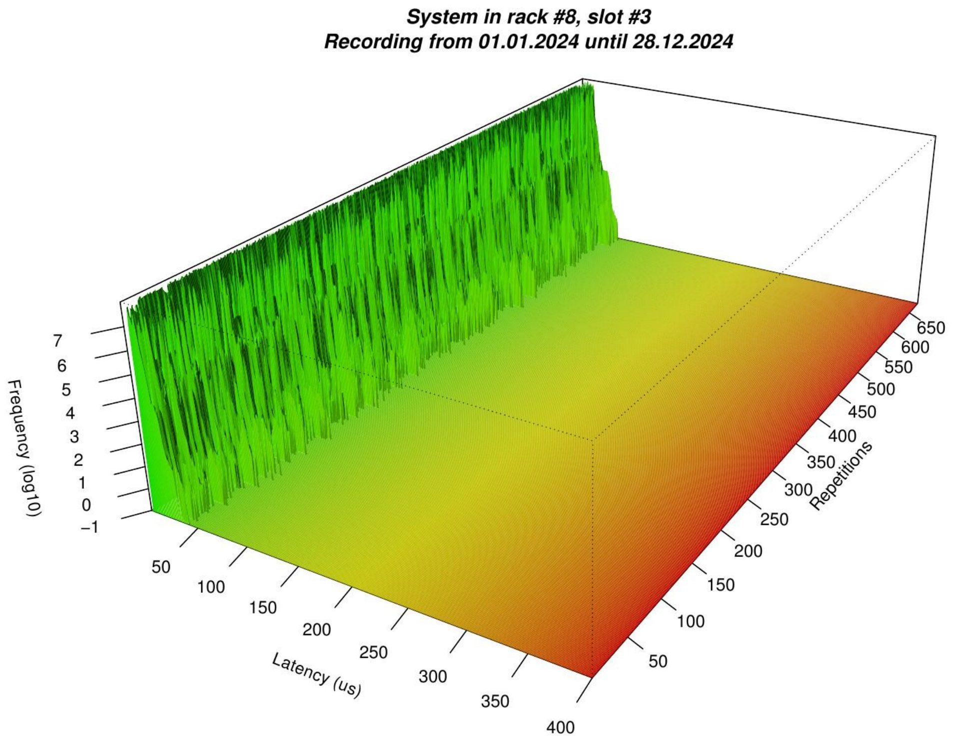
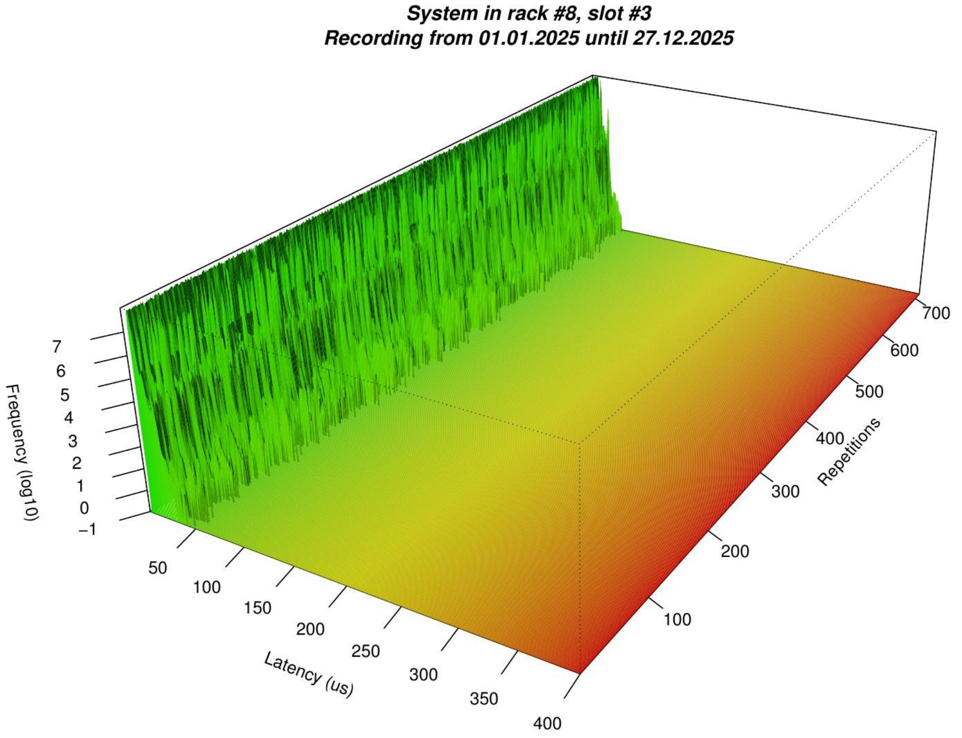
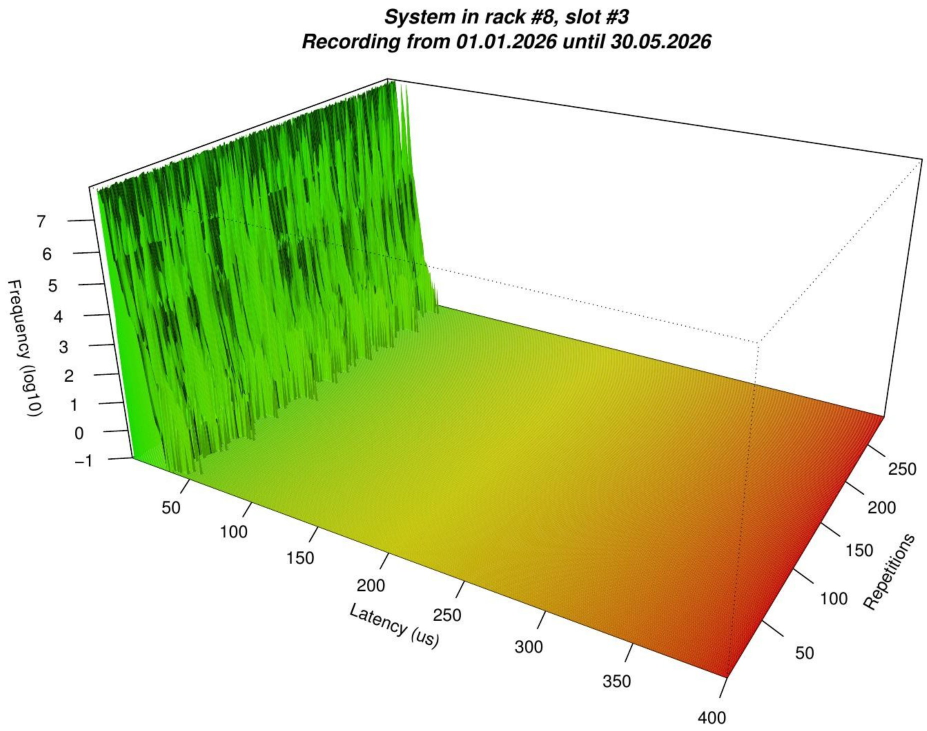


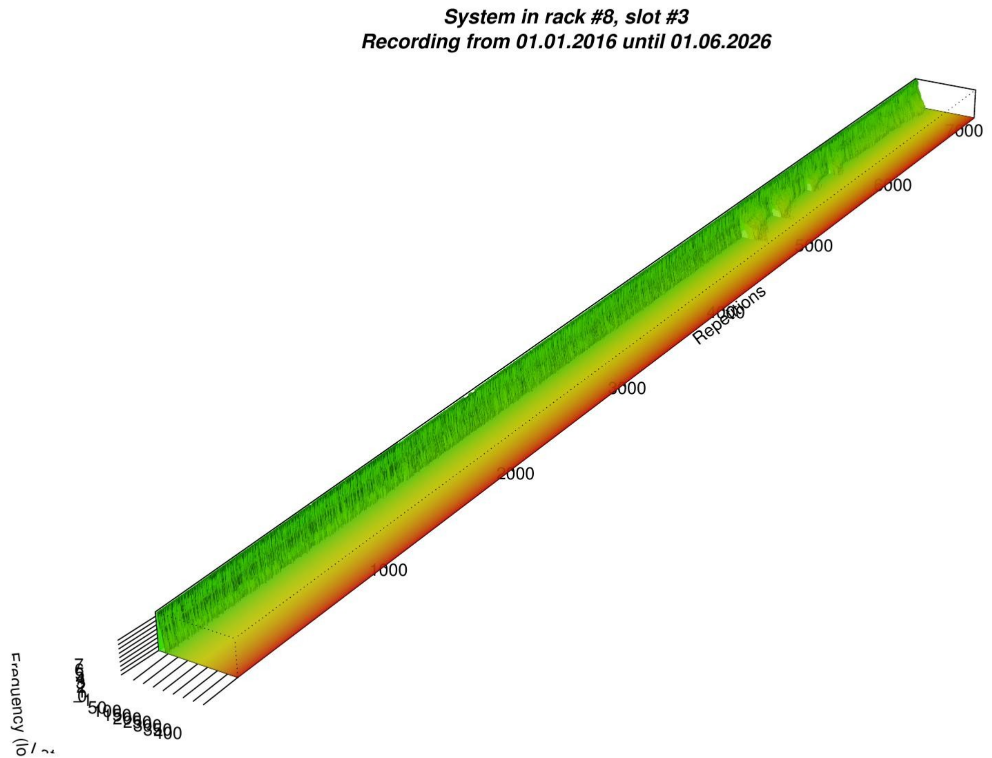
Long-term latency plots



Long-term latency plots



Long-term latency plots

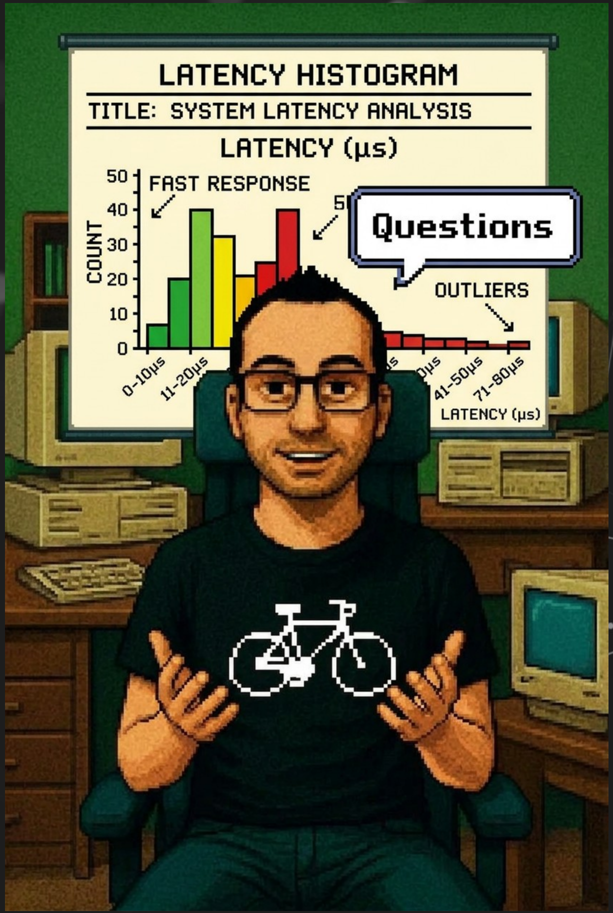


Recent activities

- Zephyr monitoring

Future work / Discussion

- Context information in long-term plots
- More Zephyr systems
- RTLA
- stress-ng



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ELISA
Enabling Linux in Safety Applications

WORKSHOP

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